

# Kinsey E. Matthews

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## Education

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- 2024-pres Doctor of Philosophy in Environmental Science and Management, University of California, Santa Barbara, Committee: Dr. Steve Gaines (co-chair), Dr. Chris Free (co-chair), and Max Czapanskiy, GPA: 4.00
- 2019-2023 Master of Science in Marine Science, Moss Landing Marine Laboratories (MLML), California State University, Monterey Bay (CSUMB). Thesis: [\*Improving species distribution models of continental shelf fishes off California\*](#). Committee: Dr. Rick Starr (co-chair), Dr. Scott Hamilton(co-chair), GPA: 4.00
- 2013-2016 Bachelor of Science in Biology, minor in Chemistry, Oregon State University. Summa Cum Laude, Honors, Dean's List, GPA: 3.90

## Professional Experience

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- 2024-present **Arnhold Fellow** – Environmental Markets Lab at UCSB  
Graduate student researcher for the Arnhold UCSB-Conservation International Climate Solutions Collaborative. Developed project on using joint dynamic species distribution models to reduce bycatch in the Eastern Tropical Pacific industrial tuna fisheries.
- 2023-2024 **California Sea Grant State Fellow** – California Fish and Game Commission  
Developed materials and engaged with various stakeholders, such as commercial fishers, environmental non-governmental organizations, and the California Department of Fish and Wildlife for projects related to fisheries bycatch, Marine Protected Areas (MPAs), aquaculture, and fishing communities. Supervisor: Melissa Miller-Henson
- 2021-2024 **Aerial Observer** – Upwell  
Assisted with aerial surveys in California, Oregon, and Washington to assess leatherback turtle and cetacean distribution and density, prey distribution and density, and oceanographic conditions, PI: Dr. Karin Forney & Dr. Scott Benson
- 2022-2023 **Research Technician** – San Jose State University Research Foundation  
Conducted weeklong surveys along the California coast, deployed and operated video systems, analyzed stationary and moving video systems, performed statistics and wrote reports and publications, managed other graduate students in the lab, PI: Dr. Rick Starr
- 2019-2022 **Student Research Assistant** – San Jose State University Research Foundation
- Conducted hook-and-line surveys inside and outside of MPAs with the California Collaborative Fisheries Research Program, processed fish and extracted otoliths and gonads, interacted with local anglers, helped run workshops for volunteer anglers and members of the Ocean Protection Council, PI: Dr. Scott Hamilton & Dr. Rick Starr
  - Compiled, organized, and analyzed logbooks and landings data to assess bycatch within California fisheries, PI: Dr. Rick Starr
  - Deployed and recovered Baited Remoted Underwater Videos and seined in the surf zone, PI: Dr. Scott Hamilton
- 2018-2019 **Lab and Field Technician** – Menge-Lubchenco lab, Oregon State University  
Researched the ecosystem dynamics and spatial distribution of rocky intertidal communities and helped to design and implement new experiments to understand the ecological consequences of sea star wasting syndrome, oversaw long term invertebrate recruitment and large predator experiments, deployed offshore moorings and Conductivity, Temperature, and Depth profilers, trained undergraduate researchers, and helped coordinate survey efforts in California, PI: Dr. Bruce Menge

- 2018 **Research Intern** – Large Marine Vertebrates Research Institute of the Philippines  
Studied the impact of provisioning on whale sharks through photo identification, monitored the behavior of green and hawksbill turtles, collaborated with the local government and community, collected data for a research publication on the use of citizen science for describing the occurrence and distribution of the ornate eagle ray, PI: Dr. Gonzo Araujo
- 2017 **Research Intern** – Bimini Biological Field Station Foundation  
Conducted telemetry and behavior studies on local elasmobranch populations, lead intern for a time placed learning study of lemon sharks, deployed and recovered baited remote underwater videos and identified fish and elasmobranch species, tagged elasmobranchs with external dart tags and Passive Integrated Transponder tags and collected fin samples, PI: Dr. Matt Smukall
- 2016 **Research Assistant** – Integrative Biology, Oregon State University  
Dissected, sanded, and imaged otoliths of juvenile rockfish, PI: Dr. Kirsten Grorud-Colvert
- 2015 **Undergraduate Animal Care Specialist** – Integrative Biology, Oregon State University  
Maintained health of sea anemone *Aiptasia* through diet preparation and tank maintenance, conducted animal husbandry experiments involving cold shock treatments between aposymbiotic and symbiotic *Aiptasia* along with bleaching experiments between different genetic strains, PI: Dr. Virginia Weiss

## Teaching Experience

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- 2025 **Graduate Teaching Assistant** – UCSB  
ESM 257 Coastal Policy and Management
- 2020 **Graduate Assistant** – MLML  
Student instructor for Scientific Writing, a graduate course with 12 students
- 2017 **Organic Chemistry Tutor** – Oregon State University  
Covered topics such as SN1, SN2, E1, E2, and alkyl halide reactions
- 2016 **Invertebrate Biology Teaching Intern** - Oregon State University  
Undergraduate instructor for higher-level invertebrate biology. Organized, prepared, and helped instruct weekly labs focused on physiology, taxonomy, and DNA barcoding and PCR analysis
- 2014-2015 **Introductory Biology Teaching Intern** – Oregon State University  
Organized, prepared, and helped instruct weekly labs focused on ecology, genetics, evolution, molecular and cellular biology, and genomics

## Honors and Awards

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- 2026-28 California Sea Grant Graduate Research Fellowship. “Connectivity of highly migratory species within West Coast National Marine Sanctuaries and implications for conservation and management.” Matthews K, Free CM, Gaines SD. \$100,000.
- 2026 Bren Conference Funding Travel Award. \$1,000
- 2025-26 H. William Kuni Bren Research Award. Matthews K, Cwikiel S, Vogel J. \$15,000
- 2025-26 Benioff Scholar in Applied Environmental Science Program. \$6,000
- 2025 UCSB Graduate Student Association Conference Travel Grant. \$300
- 2023 CSUMB nominee for Western Association of Graduate Schools' Distinguished Master's Thesis in STEM Award
- 2023 Western Groundfish Conference, 2<sup>nd</sup> place student poster presentation. \$200
- 2022 Xiphias Martin Scholarship, MLML. \$500

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| 2022    | Monterey Area Research Institutions' Network for Education Inter-Institutional Synergy Grant                                                                                 |
| 2020-22 | University Grant, CSUMB (3x)                                                                                                                                                 |
| 2021    | Council on Ocean Affairs, Science and Technology (COAST) Dr. Kenneth H. Coale Graduate Scholar Awards (Formerly known as the COAST Graduate Student Research Award), \$3,000 |
| 2020-21 | MLML Wave Awards (2x), \$1,250                                                                                                                                               |
| 2020    | COAST Student Travel Award                                                                                                                                                   |
| 2013-16 | Academic Achievement                                                                                                                                                         |
| 2014    | Academic Achievement & Excellence Award for D1 Athletes                                                                                                                      |
| 2013    | International Baccalaureate Diploma                                                                                                                                          |

### **Presentations and Talks (\*Guest lecture)**

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| 2026 | <b>K. Matthews</b> , S. Cwikiel, J. Vogel, S. Gaines, S. Stevenson, D. Crear, D. Ovando, J. Lopez. Multispecies bycatch patterns in the Eastern Tropical Pacific (ETP): towards developing near real-time bycatch avoidance tools. Ocean Sciences Meeting, Glasgow, Scotland |
| 2025 | <b>K. Matthews</b> , A. Giraldo-Ospina, J. Caselle, C. Free. The relative impacts of climate change, fisheries management, and market forces on the distribution of California fishing grounds. Talk. Western Society of Naturalists, San Diego California                   |
| 2023 | <b>K. Matthews</b> , R. Starr. Improving Species Distribution Models of Continental Shelf Fishes. Poster, Western Society of Naturalists, Monterey California.                                                                                                               |
| 2023 | <b>K. Matthews</b> , R. Starr. Improving Species Distribution Models of Continental Shelf Fishes. Poster, Western Groundfish Conference, Juneau Alaska.                                                                                                                      |
| 2022 | <b>K. Matthews</b> . Marine Protected Areas, Alisal High School. In person.*                                                                                                                                                                                                 |
| 2022 | <b>K. Matthews</b> , Grace Teranishi. Invasive Species, Alisal High School. In person.*                                                                                                                                                                                      |
| 2021 | <b>K. Matthews</b> . Marine Protected Areas, North Monterey County High School, Online.*                                                                                                                                                                                     |
| 2021 | <b>K. Matthews</b> . Life as a graduate student, Sea Lion Bowl, Online.*                                                                                                                                                                                                     |
| 2020 | <b>K. Matthews</b> , R. Starr, R. Fields, K. Cieri, J. Mohay, M. Gleason, C. Keczy, F. Cazenave. Survey like a BOSS: Use of a Stereo Video Lander to Survey Rocky Habitats in California. Talk, American Fisheries Society, Online.                                          |

### **Publications**

1. Langlois, T., C. Spencer, B. A. Gibbons, K. J. Griffin, K. Adams, C. Aston, N. Barrett, A. Bastiaansen, D. Beach, A. Bernard, T. Bond, G. R. Carey, J. E. Caselle, K. Cieri, G. H. Cummins, K. Cure, S. de Lestang, J. Fitzhardinge, A. Giraldo-Ospina, G. Grammer, D. R. Guilfoyle, C. Henderson, S. Hickey, J. Hicks, R. Hovey, C. Huveneers, D. Ierodiaconou, J. Keesing, N. Knott, J. L. Lavers, S. Lindfield, J. Lindholm, S. Mastrantonis, **K. Matthews**, M. L. Navarro, J. Partridge, D. Pelletier, C. V. H. Piggott, R. Przeslawski, B. Radford, M. Rees, R. Reynolds, F. A. Rolim, A. Smith, F. Spencer, R. Starr, S. Thompson, I. Webb, W. Webb, S. Whitmarsh, J. Williams, and J. Monk. Benthic Observation Survey System (BOSS) for surveys of marine benthic habitats. *Methods Ecol. Evol.* DOI: 10.1111/2041-210X.70010
2. **Matthews, K.**, R. Fields, K. Cieri, J. Mohay, M. Gleason, R. Starr. Stereo-video landers can rapidly assess marine fish diversity and community assemblages. *Front. Mar. Sci.* <https://doi.org/10.3389/fmars.2024.1368083>
3. **Matthews, K.**, J. Mohay, J. Todd, R. Starr. 2022. Bycatch in the California halibut (*Paralichthys californicus*) trawl fishery. *Bull. Southern California Acad. Sci.* 121(2): 88-109. <https://doi.org/10.3160/0038-3872-121.2.88>

4. Araujo, G., C. Legaspi, **K. Matthews**, A. Ponzo, A. Chin, M. Manjaji-Matsumoto B. 2020. Citizen science sheds light on the cryptic ornate eagle ray *Aetomylaeus vespertilio*, Aquatic Conserv: Mar Freshw Ecosyst. 30:10. <https://doi/10.1002/aqc.3457>

### ***In Preparation***

1. **Matthews, K.**, R. Starr. Improving species distribution models of continental shelf fishes off California. For submission to Diversity and Distributions.

## **Community Outreach & Mentorship**

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| 2026      | <b>Member</b> – Bren PhD Symposium                                                                                                                                                                                                                                                                                                                                  |
| 2025-26   | <b>PhD Advisor</b> – MESM Group Project<br>Provide advice, expertise, guidance, and help with the planning, development, and analysis of the master's group project "Evaluating Habitat suitability at Apricot Lane Farms for the Proposed Introduction of the California red-Legged Frog ( <i>Rana draytonii</i> )"                                                |
| 2025      | <b>PhD Mentor</b> – Bren-Environmental Science Fellows Program                                                                                                                                                                                                                                                                                                      |
| 2025      | <b>PhD Mentor</b> – emLab Arnhold Fellows                                                                                                                                                                                                                                                                                                                           |
| 2024-25   | <b>PhD Educator</b> – Family Ultimate Science Exploration<br>Provide educational presentations and hands-on experiments for local high schools in Santa Barbara.                                                                                                                                                                                                    |
| 2024-25   | <b>PhD Mentor</b> – Student Engagement and Enrichment in Data Science<br>Mentor undergraduate students interested in STEM disciplines and environmental science.                                                                                                                                                                                                    |
| 2024-25   | <b>PhD Fellow</b> – Bren Environmental Leadership & The Nature Conservancy<br>Mentor middle and high school students at the Jack and Laura Dangermond Preserve on day long field trips, assist and lead environmental outdoor activities.                                                                                                                           |
| 2021-2023 | <b>Program Leader &amp; Outreach Coordinator</b> – Futures Leaders of Marine Science<br>Manage, organize and coordinate lectures given by local marine graduate students to Monterey County high schools. Received the 2022 MARINE Synergy Grant for financial support.                                                                                             |
| 2020-2022 | <b>Room Leader and Score Keeper</b> – Sea Lion Bowl                                                                                                                                                                                                                                                                                                                 |
| 2020-2021 | <b>President</b> , Student Body Officers, Moss Landing Marine Laboratories<br>Main liaison between student body and faculty and staff, hosted sessions for students to voice concerns about the program, organized mental health workshops, created a COVID relief fund, hosted and facilitated staff appreciation week, hosted activities and events for students. |

## **Skills**

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| <ul style="list-style-type: none"> <li>• Microsoft Office Suite</li> <li>• Wordpress</li> <li>• R Studio</li> <li>• Species distribution modeling</li> <li>• Image J</li> </ul> | <ul style="list-style-type: none"> <li>• Esri ArcGIS Pro and arcMap</li> <li>• SeaGIS EventMeasure</li> <li>• Boating experience</li> <li>• PADI Open Water, AAUS Scientific Diver, NAUI Nitrox Diver, Rescue Diver</li> </ul> |
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